EVIDENCE-BASED HEALTH · PERFORMANCE · LONGEVITY

The Peptide Research Guide

A plain-English overview of what the research actually shows for 15 widely-discussed research peptides and growth hormone — mechanism, what's studied, evidence quality, and current FDA status. Every entry is sourced directly from PubMed.

A note on what this guide is (and isn't): This is a research reference, not a usage protocol. You will not find doses, cycle lengths, or stacking instructions here — intentionally. Evidence quality varies enormously across these compounds, and this guide aims to give you an honest read on where each one actually stands, not to tell you what to do with that information. This is for educational and research purposes only and is not medical advice. Consult a licensed physician before making any health decisions.

BPC-157

WHAT IT IS

A synthetic pentadecapeptide (15 amino acids) derived from a sequence found in human gastric juice. Proposed mechanisms include upregulation of VEGF (blood vessel formation), modulation of nitric oxide signaling, and effects that appear to support tissue and gut lining repair in animal studies.

RESEARCHED FOR

Tissue and tendon healing, gut lining repair, angiogenesis, and broader "cytoprotection" effects across injury and inflammation models.

EVIDENCE QUALITY

Extensive: 100+ published animal and in-vitro studies spanning roughly three decades. Human controlled trial data remains very limited by comparison — mostly case reports and early-stage observations, not large randomized trials.

FDA STATUS

Not FDA-approved. Sold as a research-use-only (RUO) compound; not available through licensed U.S. compounding pharmacies.

SOURCES (PubMed)

Sikiric P, et al. Cytoprotection as a Unifying Strategy for Hemorrhage and Thrombosis. Pharmaceuticals (Basel). 2026. PMID: 41901308. DOI: 10.3390/ph19030463

Matek D, et al. Tendon, Ligament, and Muscle Injury Therapy Perspectives with BPC 157. Pharmaceuticals (Basel). 2026. PMID: 41754849. DOI: 10.3390/ph19020309

TB-500 / Thymosin Beta-4

WHAT IT IS

A synthetic version of a naturally occurring peptide upregulated in injured tissue. Works primarily through actin regulation, which drives cell migration — a key step in wound and tissue repair.

RESEARCHED FOR

Corneal and skin wound healing, angiogenesis, and tissue regeneration. A related compound (RGN-259, eye-drop formulation) has gone through registered clinical trials for corneal healing specifically.

EVIDENCE QUALITY

Strong animal and mechanistic data; the corneal-specific formulation (RGN-259) has reached human clinical trials, though full FDA approval for that indication is still pending as of the most recent published research.

FDA STATUS

Not FDA-approved as a general therapeutic. The related RGN-259 eye-drop formulation has been through clinical trials but does not yet have full FDA approval.

SOURCES (PubMed)

Nguyen J, et al. Engineered Tandem Thymosin Peptide Promotes Corneal Wound Healing. Invest Ophthalmol Vis Sci. 2025. PMID: 41235866. DOI: 10.1167/iovs.66.14.31

Li W, et al. Thymosin Beta 4 Promoting Transplanted Fat Survival via ADSCs. Aesthetic Plast Surg. 2024. PMID: 38409346. DOI: 10.1007/s00266-024-03861-1

CJC-1295

WHAT IT IS

A modified analog of growth hormone-releasing hormone (GHRH) engineered to bind serum albumin, extending its half-life far beyond natural GHRH so it stimulates a longer, more sustained pulse of growth hormone release from the pituitary.

RESEARCHED FOR

Growth hormone secretion, often studied alongside GH-axis research more broadly.

EVIDENCE QUALITY

Foundational pharmacology has been published in a peer-reviewed endocrinology journal (rat and in vitro data establishing its extended half-life); large-scale human trial data for CJC-1295 specifically is limited compared to approved GH therapies.

FDA STATUS

Not FDA-approved as a standalone therapeutic. RUO status; not an approved drug product.

SOURCES (PubMed)

Jetté L, et al. hGRF(1-29)-albumin bioconjugates: identification of CJC-1295 as a long-lasting GRF analog. Endocrinology. 2005. PMID: 15817669. DOI: 10.1210/en.2004-1286

Ipamorelin

WHAT IT IS

A pentapeptide and selective growth hormone secretagogue — it stimulates GH release through the ghrelin/GHRP receptor pathway with notably less effect on cortisol and ACTH than earlier secretagogues in its class.

RESEARCHED FOR

Growth hormone release with a more selective hormonal profile than older secretagogues; studied as a candidate for conditions involving GH deficiency.

EVIDENCE QUALITY

Original pharmacology established in animal models (rat, swine) in peer-reviewed endocrinology literature. Human trial data specific to ipamorelin alone is limited.

FDA STATUS

Not FDA-approved as a standalone product.

SOURCES (PubMed)

Raun K, et al. Ipamorelin, the first selective growth hormone secretagogue. Eur J Endocrinol. 1998. PMID: 9849822. DOI: 10.1530/eje.0.1390552

Tesamorelin

WHAT IT IS

A synthetic analog of growth hormone-releasing hormone (GHRH) that stimulates the pituitary to release growth hormone, with well-documented effects on visceral fat specifically.

RESEARCHED FOR

Reduction of visceral adipose tissue, originally studied in the context of HIV-associated lipodystrophy.

EVIDENCE QUALITY

One of the most clinically established peptides on this list — multiple Phase III randomized controlled trials, plus more recent research (2024) confirming efficacy in patients on modern antiretroviral regimens.

FDA STATUS

FDA-approved (brand name Egrifta) since November 2010, specifically for reduction of excess visceral fat in HIV patients with lipodystrophy. Use outside that approved indication is off-label.

SOURCES (PubMed)

Russo SC, et al. Efficacy and safety of tesamorelin in people with HIV on integrase inhibitors. AIDS. 2024. PMID: 38905488. DOI: 10.1097/QAD.0000000000003965

Spooner LM, Olin JL. Tesamorelin: a GHRH analogue for HIV-associated lipodystrophy. Ann Pharmacother. 2012. PMID: 22298602. DOI: 10.1345/aph.1Q629

Semaglutide

WHAT IT IS

A GLP-1 receptor agonist — it mimics a gut hormone that regulates appetite and blood sugar, slowing gastric emptying and increasing satiety.

RESEARCHED FOR

Type 2 diabetes management and chronic weight management; more recently, cardiovascular risk reduction and metabolic dysfunction-associated steatohepatitis (MASH).

EVIDENCE QUALITY

Extremely well-studied: large Phase III randomized controlled trials across multiple indications, ongoing post-marketing research, and recent FDA clearance for additional indications beyond the original approval.

FDA STATUS

FDA-approved (Ozempic, Wegovy, Rybelsus) for type 2 diabetes and chronic weight management, with additional approved indications continuing to expand.

SOURCES (PubMed)

Aljabri AA. Semaglutide in Metabolic Medicine. Saudi Med J. 2026. PMID: 42237968. DOI: 10.15537/1658-3175.1060

Tirzepatide

WHAT IT IS

A dual GIP/GLP-1 receptor agonist — the added GIP mechanism (compared to semaglutide) appears to produce a synergistic effect on fat metabolism and energy expenditure.

RESEARCHED FOR

Type 2 diabetes and chronic weight management; head-to-head trials show greater average weight loss than semaglutide in matched populations.

EVIDENCE QUALITY

Extensive Phase III trial data (SURPASS and SURMOUNT programs), including the SURMOUNT-5 head-to-head trial directly comparing it to semaglutide.

FDA STATUS

FDA-approved (Mounjaro for type 2 diabetes, Zepbound for chronic weight management).

SOURCES (PubMed)

Brož J, et al. Tirzepatide. Ceska Slov Farm. 2026. PMID: 42334912. DOI: 10.36290/csf.2026.020

Retatrutide

WHAT IT IS

A triple hormone receptor agonist acting on GLP-1, GIP, and glucagon receptors simultaneously — the added glucagon target is unique among this class of compounds.

RESEARCHED FOR

Weight management and type 2 diabetes; the largest Phase III trial (TRIUMPH-1) reported up to 30% body weight loss over two years at the highest dose in participants with severe obesity.

EVIDENCE QUALITY

Multiple large Phase III trials have reported positive topline results in 2025 and 2026 (TRIUMPH-4, TRANSCEND-T2D-1, TRIUMPH-1), though as of this guide full peer-reviewed publication of some of these results is still working through the publication process.

FDA STATUS

Not yet FDA-approved for any indication. Fully investigational, still in the Phase III trial process with Eli Lilly.

SOURCES (PubMed)

Topline trial results reported via company press releases and presented at medical conferences (e.g., TRIUMPH-1, TRIUMPH-4, TRANSCEND-T2D-1); full peer-reviewed publications are pending as of this guide's writing. Check ClinicalTrials.gov for the most current registered trial data.

MOTS-c

WHAT IT IS

A mitochondrial-derived peptide — encoded within mitochondrial DNA rather than the nuclear genome. Functions through AMPK and PGC-1α signaling pathways involved in cellular energy regulation.

RESEARCHED FOR

Metabolic regulation, mitochondrial bioenergetics, and insulin sensitivity; often discussed as an “exercise mimetic” candidate.

EVIDENCE QUALITY

Growing animal and mechanistic research base, including a 2026 study showing improved muscle mitochondrial bioenergetics in mouse models. Human data remains early-stage.

FDA STATUS

Not FDA-approved. RUO status.

SOURCES (PubMed)

Gudiksen A, et al. MOTS-c improves intrinsic muscle mitochondrial bioenergetic health. Free Radic Biol Med. 2026. PMID: 41520850. DOI: 10.1016/j.freeradbiomed.2026.01.002

Tero-Vescan A, et al. Exercise-Induced Muscle-Fat Crosstalk. Pharmaceuticals (Basel). 2025. PMID: 40872612. DOI: 10.3390/ph18081222

Epitalon (Epithalon)

WHAT IT IS

A synthetic tetrapeptide (Ala-Glu-Asp-Gly) based on a bovine pineal gland extract. Proposed effects include influence on melatonin synthesis and telomerase enzyme activity.

RESEARCHED FOR

Geroprotective and neuroendocrine effects, circadian regulation, antioxidant and antimutagenic activity.

EVIDENCE QUALITY

A 2025 comprehensive review notes roughly 25 years of in vitro, in vivo, and in silico research, but also highlights that physico-chemical and structural investigation remains limited, and much foundational work (largely from earlier Russian research) hasn't been replicated under modern trial standards.

FDA STATUS

Not FDA-approved. RUO status.

SOURCES (PubMed)

Araj SK, et al. Overview of Epitalon — Highly Bioactive Pineal Tetrapeptide. Int J Mol Sci. 2025. PMID: 40141333. DOI: 10.3390/ijms26062691

SS-31 / Elamipretide

WHAT IT IS

A mitochondria-targeted peptide that selectively binds cardiolipin on the inner mitochondrial membrane, supporting electron transport chain efficiency and ATP production.

RESEARCHED FOR

Mitochondrial dysfunction in age-related and rare diseases; has been studied in specific cardiac and mitochondrial disease contexts rather than general longevity use.

EVIDENCE QUALITY

Solid preclinical mechanistic base, and among the more clinically advanced peptides in the mitochondrial-targeting category — it has reached human clinical trials for specific conditions, though its studied population is narrower than general aging use.

FDA STATUS

Not FDA-approved for general use. Has been studied in clinical trials for specific mitochondrial and cardiac conditions.

SOURCES (PubMed)

Szeto HH. Stealth Peptides Target Cellular Powerhouses to Fight Age-Related Diseases. Protein Pept Lett. 2018. PMID: 30381054. DOI: 10.2174/0929866525666181101105209

PT-141 (Bremelanotide)

WHAT IT IS

A melanocortin receptor agonist (MC4R) that acts on brain pathways involved in sexual arousal, distinct from drugs that work through vascular mechanisms.

RESEARCHED FOR

Hypoactive sexual desire disorder (HSDD) in women.

EVIDENCE QUALITY

Strong: multiple Phase II and Phase III randomized controlled trials supporting FDA approval, with published data on efficacy and the most common side effects (nausea, facial flushing, headache).

FDA STATUS

FDA-approved (brand name Vyleesi) for premenopausal women with acquired, generalized HSDD.

SOURCES (PubMed)

Mayer D, Lynch SE. Bremelanotide: New Drug Approved for Treating HSDD. Ann Pharmacother. 2020. PMID: 31893927. DOI: 10.1177/1060028019899152

GHK-Cu (Copper Peptide)

WHAT IT IS

A naturally occurring copper-binding tripeptide (glycyl-L-histidyl-L-lysine) present in human plasma that declines with age. Stimulates collagen synthesis and modulates extracellular matrix remodeling.

RESEARCHED FOR

Skin regeneration, wound healing, and anti-aging skincare applications; also studied for broader tissue repair and gene expression effects.

EVIDENCE QUALITY

A substantial and long-running research base, including a widely cited 2015 review describing effects across skin, hair follicles, and wound models in multiple species, plus cell-culture studies on keratinocyte and stem cell activity.

FDA STATUS

Not FDA-approved as a drug; used in cosmetic products, which are regulated differently than drugs (cosmetic claims, not therapeutic claims).

SOURCES (PubMed)

Pickart L, et al. GHK Peptide as a Natural Modulator of Multiple Cellular Pathways in Skin Regeneration. Biomed Res Int. 2015. PMID: 26236730. DOI: 10.1155/2015/648108

Choi HR, et al. Stem cell recovering effect of copper-free GHK in skin. J Pept Sci. 2012. PMID: 23019153. DOI: 10.1002/psc.2455

Thymosin Alpha-1

WHAT IT IS

An immune-modulating peptide derived from the thymus gland, historically used to support immune function in various clinical contexts.

RESEARCHED FOR

Immune modulation in the context of infections (including COVID-19), autoimmune conditions, and as an adjunct in some cancer treatment protocols.

EVIDENCE QUALITY

A 2024 narrative review found over 11,000 human subjects across 30+ trials worldwide, reporting consistent safety and effectiveness signals — notably, this review specifically pushed back on the FDA's 2023 regulatory restriction of this peptide (along with 21 others), arguing the restriction wasn't supported by the accumulated clinical evidence.

FDA STATUS

Restricted by the FDA in 2023 (placed on a list of bulk substances not eligible for compounding). Not FDA-approved as a standalone drug in the U.S., though it holds approvals in some other countries under the brand name Zadaxin.

SOURCES (PubMed)

Dinetz E, Lee E. Comprehensive Review of the Safety and Efficacy of Thymosin Alpha 1 in Human Clinical Trials. Altern Ther Health Med. 2024. PMID: 38308608.

HGH (Human Growth Hormone / Somatropin)

WHAT IT IS

The recombinant form of the body's own growth hormone, produced in the pituitary gland. Regulates growth, cell repair, and metabolism throughout the body, primarily via IGF-1.

RESEARCHED FOR

Growth hormone deficiency (pediatric and adult), certain growth disorders, and HIV-related wasting. Widely discussed off-label for anti-aging and body composition purposes.

EVIDENCE QUALITY

One of the most extensively studied hormones in medicine, with decades of clinical trial data for its approved indications. Evidence for anti-aging or performance use outside diagnosed deficiency is far thinner and is not what the approval or majority of trial data actually supports.

FDA STATUS

FDA-approved for multiple specific indications (growth hormone deficiency, certain pediatric growth disorders, HIV wasting). Newer long-acting weekly formulations have also received FDA approval in recent years. Use for general anti-aging or performance purposes in the absence of diagnosed deficiency is off-label and not an FDA-approved use.

SOURCES (PubMed)

Grimberg A, Hawkes CP. Growth Hormone Treatment for Non-GHD Disorders. J Clin Endocrinol Metab. 2024. PMID: 37450564. DOI: 10.1210/clinem/dgad417

Miller BS. What do we do now that the long-acting growth hormone is here? Front Endocrinol. 2022. PMID: 36072938. DOI: 10.3389/fendo.2022.980979

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